451/2
GEOGRAPHY PAPER 1
MARKING SCHEME

1a) Give three examples of mechanically formed sedimentary rocks (3mks)

- Clay stone
- Siltstone –shale
- Mudstone

b) state two changes that occur in sedimentary rocks when they are subjected to heat and pressure

- New minerals are formed
- Minerals recrystalize further
- Rock particle s become compacted
- The physical appearance of the rock changes
- Metarmophism without any details (Any 2x1 = 2 marks)

2. a) i) Name the latitude marked G

Tropic of cancer (1mk)

ii) what is the angle of inclination of the earth axis from its orbit ? (1mk)

a) State three feautures that make the earth planet different from there planets in the solar system

- Deposition /by water /ice
- Erosion/by wind /ice
- Human activities /damming /blowing up of land with explosives
- Mass movement

b) State three ways in which lakes influence the natural environment (3mks)

- Are reservoirs in the water cycle
- Support bio-diversity/support floras and fauna
- Enable self- purification of water and air
- Modify local weather and climate
- Regulation of river /flow/controlling flooding

4. In your answer booklet ,name the air current marked E &F (2mks)

E- sea /lake breeze
F – land breeze

a) Give the reason why it cools as it rises (2mks)

- As it rises it expands thus spreading over a wider area and hence becoming cooler

5, a) Name the features marked X and Y (2mks)

X - a cave
Y – a blowhole

b) State three conditions necessary for the formation of a sand spit

- A (weak) long current /drift
- An indented coastline /presence of a headland
- A shallow continental shelf

SECTION B

a) Contours

b) trigonometrically station

ii) Adjoining map sheet 120l$_4$ – Nyeri

Adjoining Map Sheet - 135/2 – Embu

iii) ratio scale 1:50,000

statement scale : 1cm rep 500m

or 1cm rep 0.5km

(1x1= 1m)

iv) area of the forest

fuel squares – 8

half squares – 40/2= 20

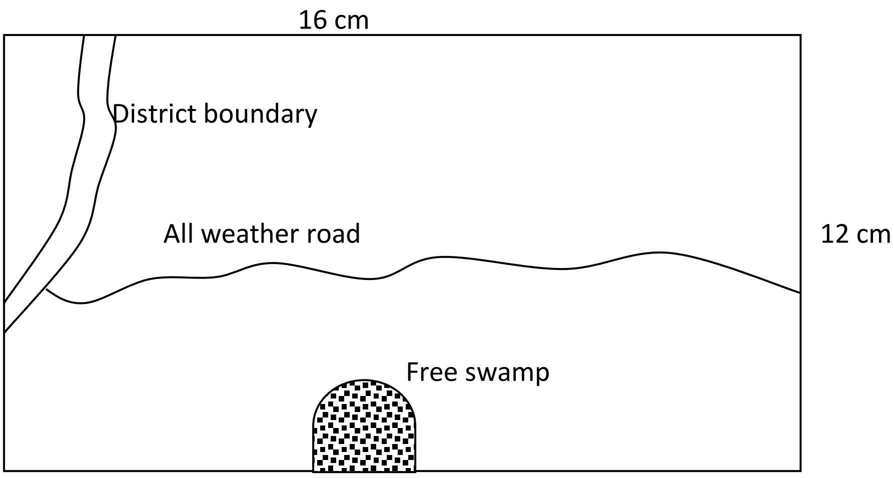
= (8+20)

Area = 28 km ^{2} (2mks)

c) two human features found in grid squares 8052

- dispensary
- church
- polytechnic
- power station
- main track/motor bale track (2x1 = 2mks)

b)



d) describe the chromate by the image

- there are many rivers in the area covered by the map
- most of the rivers are permanent e.g River Thego
- the rivers originate /flow from Mt Kenya forest southwards
- the rivers and the tributaries form enclitic drainage pattern
- there is a dam st the lower course of R, thego

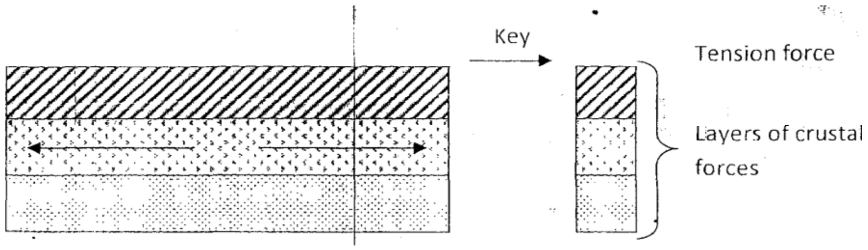
e) citing evidence from the map explain three factors that influence the distribution of settlement in the area covered by the map

- the nature of ratio : steep slopes evidenced by close contours /on Mt Kenya forests have nil or few settlements because it is difficult to construct
- forests areas with thick forests to the north /Mt Kenya have no settlement because they have been gazette /set aside for the forests
- swamps areas south eastern (tree swamp) have no settlement for fear of diseases//difficult to construct houses

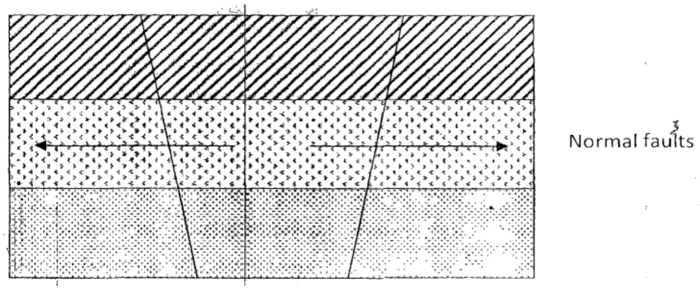
- river valleys have few /nil settlement e.g along R. sagana because of fear of flooding /water born diseases
- Market/shopping centers e.g Karatina have dense settlement attracted by social amenities/commercial activities

7. a)(i) tilt block
  - Escarpment/scrap slope
  - Block mountain/horsts *Any 2x1= 2mks*

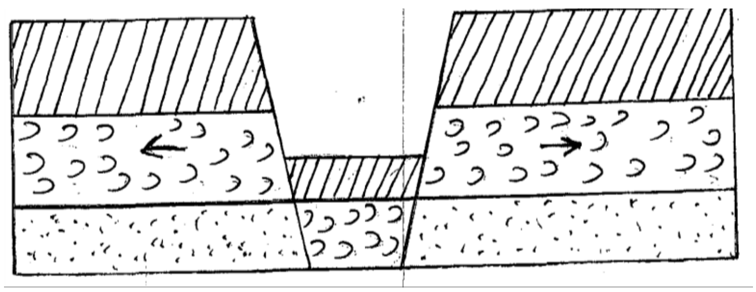
(ii)



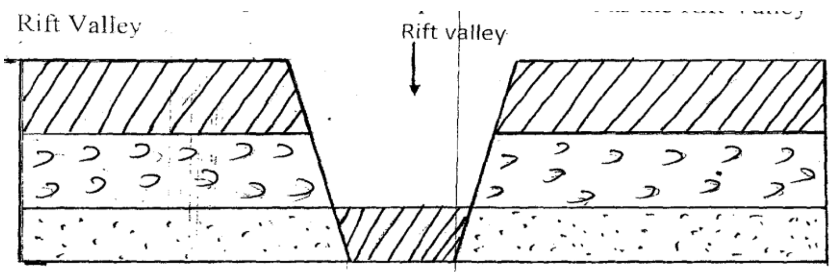
- Layers of rocks are subjected to tensional forces when there is some tensional forces when there is some instability within the earth's crust
- Parallel normal faults develop /lines of weakness develop



The middle part gradually sinks/subsides



The sunken middle part forms a depression known as the Rift valley



b) Faulting/scrap metals make it difficult to construct roads or railways
- Depression in the Rift Valley contain water that forms lakes
- Faulting exposes minerals such as diatomite

- Step faulting makes rivers to have water falls ,rapids and cataracts
- The scrap slopes/steep slopes tend to discourage settlement
- Some rivers such as Katonga in Uganda had their directions of flow changed (Any 4x2 = 8mks)

c) To enable them draw up study objectives /hypothesis
- To familiarize themselves with the area of study
- To enable them draw a route map
- To enable them a work schedule/plan activities
- To enable them identify/sort our relevant tools /equipment for the study
- To identify suitable methods of data collection
- To seek permission from the occupants of their site of study
- To enable them prepare financial (Any x1 =4 mks)

d) -It is expensive
-It is time consuming
- it is tiresome
- it is limited only to direct sources/primary sources
- it is only suitable to the signed people

8 a) i) what is glacier
- A mass of ice of limited with which moves outwards from a central area of ice accumulation (1x1)= 1mk

ii) differences between glacier trough and river valley

GLACIER TROUGH	RIVER VALLEY
u- shaped	v- shaped
Formed by glacier erosion	Formed by river erosion
Have formulated spurs	Has inter-locking spurs
The course is straight	The course is mean /zig- zag

b) Two ice ages
- Pre-colonial C 5000 million years ago
- Ordinician C 500 million years ago
- Carboniferous/Permian (350-280 million years ago
- Pleistocene C 1 million years ago 2x1 = 2mks

ii) Two ways in which we use plastic flowage
- A great pressure is exerted on the layers at the bottom sides and center within the ice mass
- This pressure causes ice particles to melt
- The melting causes some particles to change their position by slightly moving downhill later than freeze again and their action which goes on all the time cause the ice to move down slope

Basil slip
- The weight of the ice cause ice layers which is in contact with the wall to melt
- This crates a film of water which as a lubricant between the ice and the surface
- To ship and slide over the underlying work

Extension flow
- It mainly affects the ice sheets in lowland areas
- When ice accumulates ,it builds up to great thickens at the center
- The resultant weight compresses the layers of ice beneath forcing them to spread out to where there is less pressure (2x1 = 2mks)

C) start the characteristics of a pyramidal peak

- It has steep sides
- It is surrounded by circles (2x1)+2mks

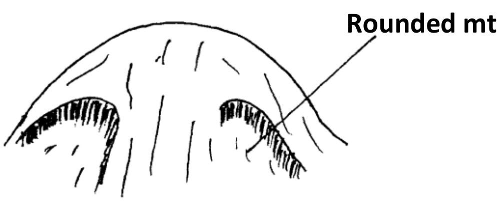
Ii) three types of movements

- Ground movement
- Medial movement
- Lateral movement
- Terminal movement

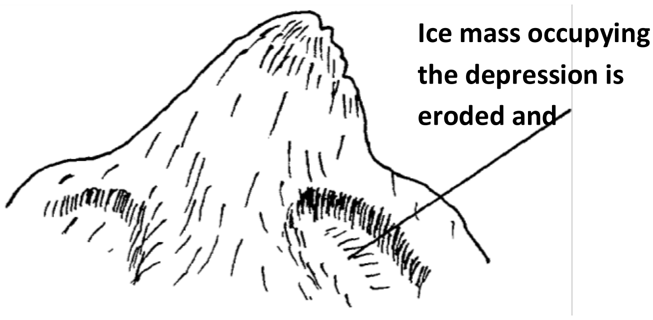
d) Describe the pressures involved in the formation f carrie lake

- Snow accumulates in a pre-existing depression on a mountain side
- The snow gets converted into ice forming a cirque glacier
- Frost action /alternating free- than action enlarges the hollow
- Abrasion /snowing action at the bottom f the glacier deepens the hoolow
- Phorching process steepens the rocks
- Eventually a deep arm- chair shaped depression known as a carrie is formed
- When water accumulates in the corrie it becomes correie lake

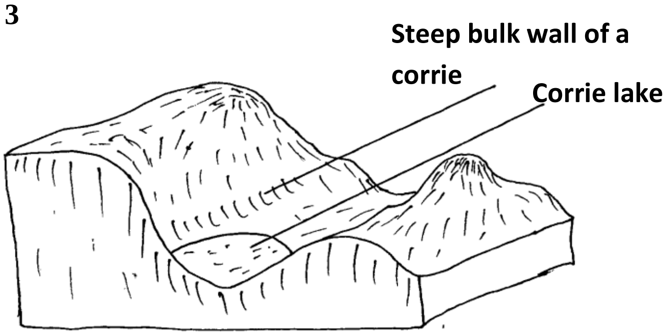
Stage 1



Stage 2



Stage 3



e) Outline four ways in which glaciated land scape is of negative significance to human activities

- Boulder clay deposits marshy landscape which harbpours wild animals
- Land become unsuitable for agriculture due to infertile deposits of sand
- Numerous lake s reduce the land available for human settlement
- It results in a rugged landscape that discourages settlement and agriculture

9 a) hydraulic action

❖ Water is forced into cracks on the river banks /water hit banks
❖ Air in the crack is compressed
❖ As the wtare retreats, pressure in the cracks is suddenly released
❖ The compreesion and widening of the cracks repeatedly
❖ The retreating water carries awy the loose particles

- The force of the moving water and the eddying effect
- Sweep away loose materials I the river channel

II) abrasion

- River carries sand ,gravel and boulders
- The load is used as a tool for scoring the load is hurled by the water against by the river waateragaimnst the banks and drafsted along river bed
- The load chips off rock on the bank and the floor (the size of the load determines the rate of erosion)
- The load of being dragged smoothens the river bend
- Eddy currents rotate rock particles in hollow and widen them in potholes

b)Local uplift of land (dynamic rejuvenation) lead to a change in the base level hence the river revives its erosive activities

- Lowering at the sea level (eustactic rejuvenation) creates shrp breaks /knick points at the river mouth . this leads to revived erosion
- Increase in discharge raise the volume of a river thus increasing its erosive power
- Presence of a hard rock out crop along the river causes breaks over which a river drops in falls and renews its erosive work

ii) river rapture may occur by headward extension of the long profile

- This happens when rivers are sharing a watershed
- The actively eroding river gradually cuts back its slope head until it encroaches upon the the divide or water shed of the other river
- Eventually the power reaches the source of the weaker river and diverts its water into its chanell
- River rapture may also ocuur where there are two adjacent rivers
- One of the rivers has more erosive power than the other
- The more powerfull river erodes away the ridge that separates the two by head ward erosion
- Eventually it encroaches into the valley of the weaker river diverting its waters into its valleys

c) Some student carried out a field study on the features along the a river

i) List three features formed as a result of river erosion

- V-shaped valley/canyons
- Pothole
- Interlocking spurs
- Waterfall rapids
- Ii) state three methods that student may have used to record their data
- Taking photographs
- Note making
- Filling questionnaires
- Making sketch digrams

ii) Significance of features resulting from deposition to human activities

- Flood plains formed by finer deposition from fertile soil for agriculture
- Raised river beds /deformed streams caused flooding which lead to loss of life nd destruction of properties
- Ox- bow lakes etc attracts tourists
- Sand deposits are extracted - provide sand for building

Any firs 2x2 = 4 mks

10 a) weathering is the disintergration of rocks in the earth;s crust while denodration reffres to all activities that wear away the earth;s crust

iii) Thre factors that influence the rate of wethareing

- Vegetation cover

- Gradient of land /slope
- Climate /rainfall and tempratuer
- Human activities /blasting/quarrying. 3x1 = 3mks

b) explain the how the following process of weatherinfg occur in arid ares
crystal growth - occur in hot desserts wher vaporation exeeds the amount of rainfall reerved

- It involves the growth of salt crystals within a rock

Capillary action drains the wter from the rock to the surface

- Once the water reaches the surface , it evaporates leaving a precipitate which crystalises n
- The crystals exten t pressure on the crevices amd widen them
- This leads to the rocks forming hollows and breaking them 2x1 = 2mks

Exploiation

- Occurs in rocks with a uniform structure /homogenous
- During the day the rock's surface heat up rapidly because of high temperature
- The rock surface expands more than the inner parts leading to cracks in the rocks
- - when temperature fall of right the surface layer of the rock contracts and more rocks develop
- The alternate expansion and contraction resulting in the peling off the rock forming a rounde mould of rock called exploiaition dome

Block disintergration

Caused by large withdrawal of temperatures
Occurs where rocks rae well formed
During the day ,the rock heats up and the expand
The repeated alternate expansion an contraction results in breakage of rocks along their joints into smaller blocks
2x7 = 2mks

Granular disintergration
Occurs in the rocks that are made of different minerals minerals (hetegrenonol)
Different minerals have different roles of expansion and contraction when exposed to alternate heating and cooling
This leads to the breaking of a rock into individual smaller granulatrt fragments and coarse grains

Ci) apart from soil creep ,name three other types of slow –massd wsting

- Talurs creep
- Solifunction
- Rock creep

3x1 =3 mks

Iii0three cause of soil creep

- Tertomic movements like earthquakes
- Plonghinh down slopes
- Splash /rain dops
- Quaryying

3x1 = 3 mks

d)i) ways in which they prepare for the field study
- formulation of objectives and hypothesis
- preparing questionnaires
- decide on methods of collecting and recordind data
- dividethe class into groups

- carrying out a pre-visit
-Aseemble the necessary tools /materials
-Design a work schedule